

1 Abstract

This report presents the implementation and simulation of a Vienna Rectifier using a balanced three-phase AC supply with an input current conditioning circuit. The rectifier consists of upper and lower diode branches and an eight-switch network, operated in a simplified mode where all switches are held on throughout the simulation. Interleaved boost legs are employed to reduce current ripple, improve EMI performance, and decrease output-filter requirements. The system drives a highly inductive load. Simulation results show that each input phase has a peak value of 311.127 V (220 V_{rms}), while the output voltage stabilizes at approximately 452.6–402 V with a switching-induced ripple of 50 V. The output current rises smoothly to 429–430.5 A, demonstrating stable DC-link operation and effective power transfer. Analysis of the diode waveforms confirms correct full-wave rectification, with the positive and negative branches conducting complementary half-cycles. Overall, the results indicate that the Vienna Rectifier delivers high-quality DC power suitable for fast-charging and other high-power conversion applications.

2 Introduction

Global transportation produces almost one-third of worldwide greenhouse gas emissions, making electric vehicles an increasingly important cleaner alternative [1], [2]. Despite economic uncertainties, global EV adoption has surpassed ten million units, supported by falling lithium-ion battery costs and the steady expansion of fast-charging networks [3]. However, large-scale deployment still faces challenges such as high battery prices, limited driving ranges, long charging durations, and significant grid issues, including harmonics and unmanaged load growth. While recent studies discuss progress in EV technology and charging infrastructure, comprehensive evaluations of power-converter topologies and control methods—particularly for fast and multiport chargers—are still relatively scarce [4], [5]. This work focuses on analyzing the Vienna Rectifier ac–dc converter architecture for next-generation fast-charging applications.

3 Analysis

3.1 Viennas Rectifier

Vienna Rectifiers (VRs) [6] improve conversion efficiency, reduce electromagnetic noise, and simplify control relative to six-switch TPBRs [7]. Interleaved implementations further lower input current ripple and enhance reliability — for example, dual-channel interleaved Vienna designs have demonstrated efficiencies above 99%. Advanced variants, including multilevel and modified Vienna topologies, enhance power capability and grid performance through improved voltage balancing and reduced switching stress [8]. A principal limitation of the standard Vienna topology is its inherently unidirectional power flow; enabling bidirectional operation typically requires substantially increased circuit complexity.

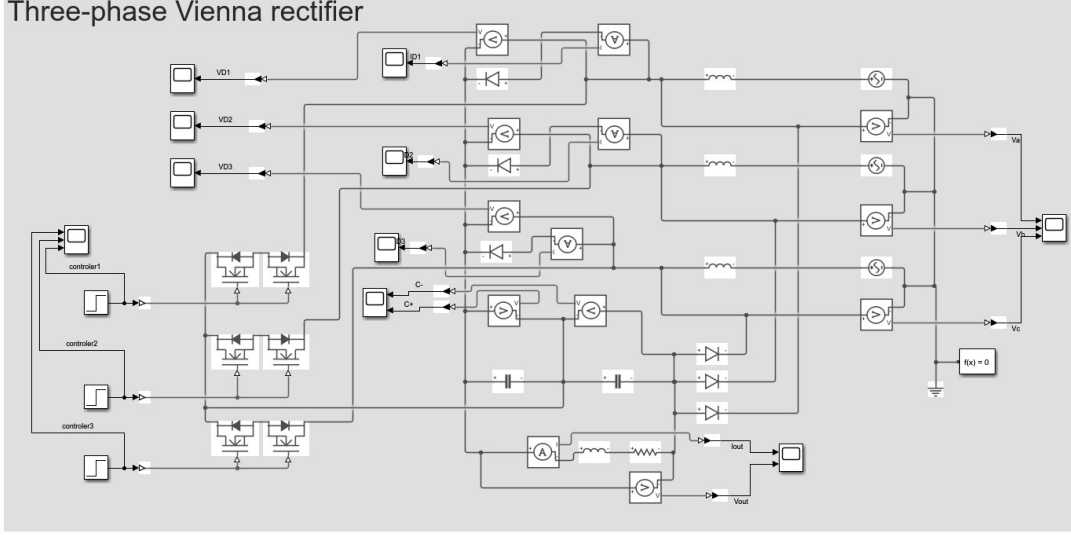


Figure 1: Vienna Rectifier Schematic

Figure 1 shows the implemented three-phase Vienna Rectifier. The system is supplied by a balanced three-phase AC input, and each phase includes a series boost inductor ($L = 1 \text{ mH}$) for input-current shaping and line-filtering. The rectifier core consists of the standard Vienna topology: a diode matrix that forms the three-level structure and one controlled bidirectional switch per phase, used to regulate the midpoint potential and achieve balanced, low-distortion input currents. The split DC-link is formed by two capacitors ($C_1 = C_2 = 1 \text{ } \mu\text{F}$), providing the three-level output and stabilizing the midpoint node.

For simplicity in this model, the switches are either held in a fixed conduction state or driven by basic control inputs without implementing a full PWM strategy. The output of the rectifier is connected to a highly inductive RL load ($L = 8 \text{ H}$, $R = 1 \text{ } \Omega$) to emulate a heavy DC-bus or charging system. Voltage and current sensors throughout the model allow monitoring of diode conduction sequences, midpoint-balancing behavior, and overall DC-link quality. All diodes and semiconductor switches include their inherent non-ideal conduction characteristics.

3.2 Output Waveforms

Figure 2a presents the balanced three-phase input voltages supplied to the Vienna Rectifier, each phase separated by 120° . The phase voltages V_a , V_b , and V_c reach a peak of 311.127 V , corresponding to $V_{\text{rms}} = 220 \text{ V}$. The resulting DC output is illustrated in Figure 2b. The output voltage rises from zero and stabilizes around $452.6\text{--}402 \text{ V}$, exhibiting a switching-induced ripple of approximately 50 V . Similarly, the output current increases smoothly from zero to nearly $429\text{--}430.5 \text{ A}$, representing normal charging behavior under load. The gradual settling of both V_{out} and I_{out} confirms stable DC-link regulation and effective power transfer through the Vienna Rectifier.

Figure 6 illustrates the internal rectification mechanism of the Vienna topology. In the conduction path shown in Figure 3a, the corresponding diode conducts during the appropriate half-cycle, generating narrow current pulses and displaying a reverse-blocking

voltage when not conducting. Conversely, the complementary conduction path illustrated in Figure 3b carries current during the opposing half-cycle, producing similar pulse characteristics and switching-related voltage ripple. The coordinated operation of these diode branches ensures that each half-cycle of the three-phase input contributes to the DC output, demonstrating correct rectification behavior and proper interaction between the diodes, midpoint control, and switching stages inherent to the Vienna Rectifier.

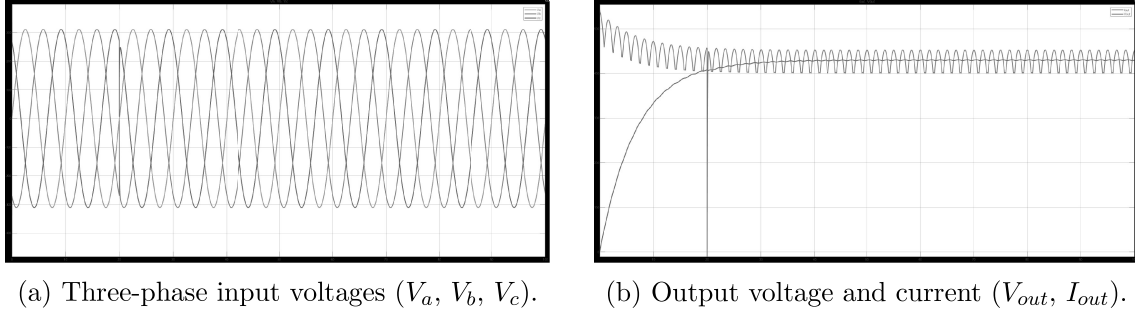


Figure 2: Input and output waveforms of the Vienna Rectifier.

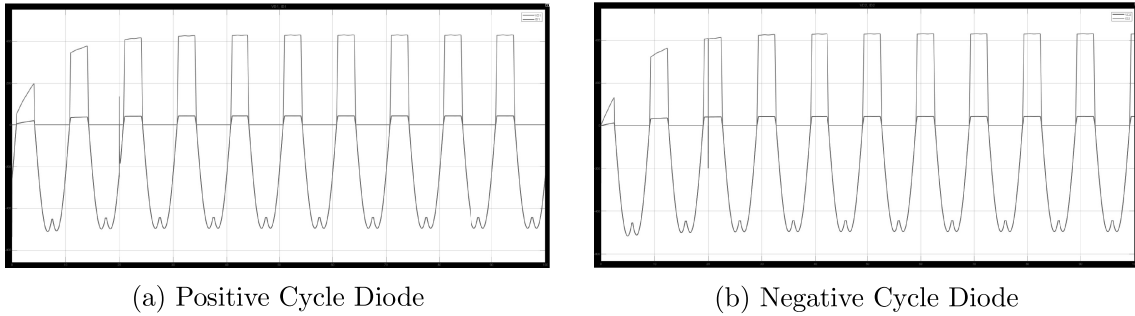


Figure 3: Voltage and Current across the Devices

3.3 Different α study case

With the Vienna Rectifier updated to use thyristor-based switches instead of passive diodes (Figure 4), the converter operation becomes dependent on the firing angle α , allowing controlled regulation of power flow and DC-link behavior. When $\alpha = 0^\circ$, the thyristors conduct immediately at the beginning of each phase voltage cycle, reproducing the same behavior as a diode-based rectifier. In this mode, the output voltage V_o follows a smooth, high-average DC level, and the output current I_o rises steadily under the inductive load, as illustrated in Figure 5a. The updated schematic diagram highlights the replacement of the diode branches with controlled thyristor devices, while maintaining the free-wheeling diode path on the DC side to ensure continuous current flow.

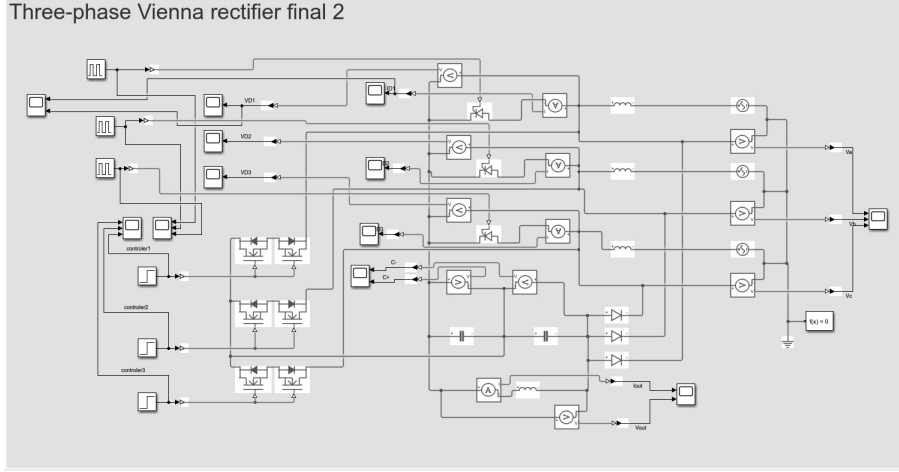


Figure 4: Vienna Rectifier with Thyristors Schematic

When the firing angle is increased to $\alpha = 60^\circ$, corresponding to the critical conduction point, the conduction interval of each thyristor is reduced. As a result, the average output voltage decreases significantly, and the current waveform exhibits deeper ripple due to shorter conduction periods. This behavior is clearly visible in Figure 5b, where both waveforms show reduced magnitude and increased distortion. Finally, at $\alpha = 100^\circ$, the thyristors delay conduction further into the voltage cycle, causing the output voltage to drop even more and forcing the free-wheeling diode to sustain current flow for a larger portion of the cycle. Figure 5c demonstrate this effect, with the current remaining continuous mainly due to the free-wheeling path, despite minimal thyristor conduction. Overall, these results confirm that introducing thyristor control provides a clear mechanism for regulating output power while highlighting the essential role of the free-wheeling diode in maintaining stable current under highly delayed firing conditions.

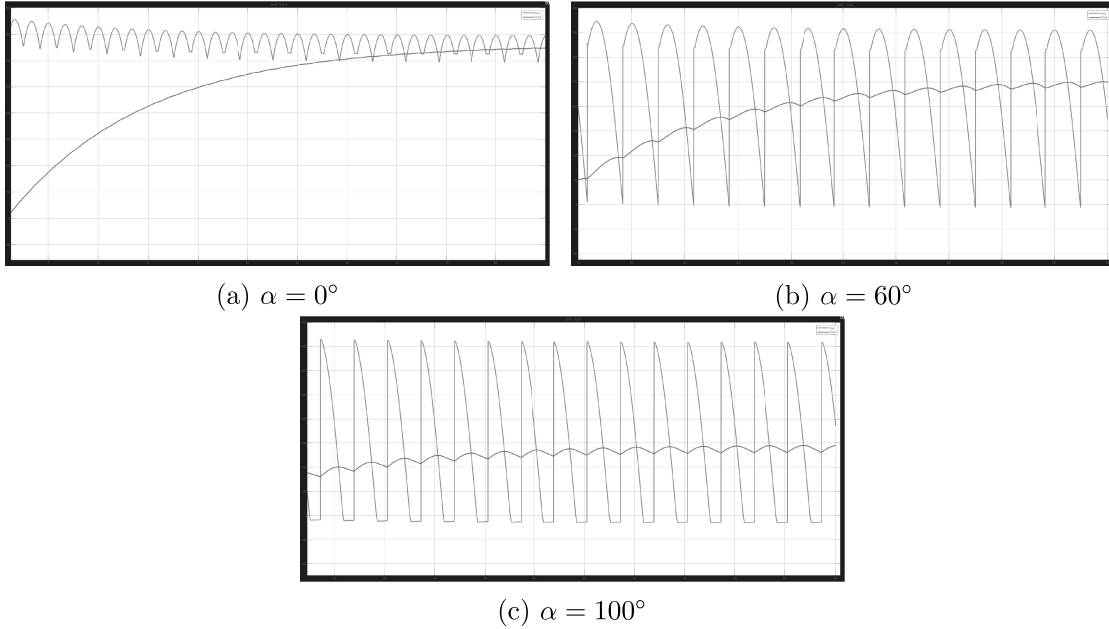


Figure 5: Output Voltage and Current for different α

3.4 Different Loads study case

At $\alpha = 100^\circ$ the Vienna rectifier behavior is strongly influenced by the load dynamics. Recall that the highly inductive load used earlier was an RL load with $R = 1 \Omega$ and $L = 8 \text{ H}$; this large inductance forces nearly continuous current flow and serves as the baseline “continuous-inductive” (CI) condition. Figure 6a shows the measured/simulated V_O and I_O for this CI case: the current remains continuous due to the large stored energy, the thyristor conduction windows are short or even absent for many cycles, and the free-wheeling path carries most of the current between delayed firing instants, producing a clamped or plateaued V_O during those intervals.

To highlight the transition to discontinuous conduction, a second test uses reduced inductance (the “discontinuous-inductive” or DCI case). In this condition (Figure 6b), the inductor can no longer sustain uninterrupted current throughout the line cycle, so I_O falls to zero for portions of each period. The waveforms show distinct current pulses that begin when a thyristor or the free-wheeling path becomes available and end when the stored energy is depleted. Correspondingly, V_O exhibits sharper transitions and larger ripple when current is absent, and the rectifier’s midpoint balancing and harmonic content become more sensitive to gating timing.

Finally, for a purely resistive load (R case) the output current follows the instantaneous rectified voltage and conduction becomes highly pulsed at $\alpha = 100^\circ$ (Figure 6c). With no inductive energy storage, thyristors conduct only during brief firing windows and the free-wheeling diode has little opportunity to sustain current. As a result, V_O and I_O show pronounced pulses and large di/dt, increasing switching and conduction stress as well as harmonic emission.

These three figures (6a, 6b, and 6c) collectively show how, at large firing delay ($\alpha = 100^\circ$), the energy stored in the load determines whether current remains continuous—reducing distortion—or becomes discontinuous and pulsed, increasing ripple, harmonics, and midpoint-control challenges.

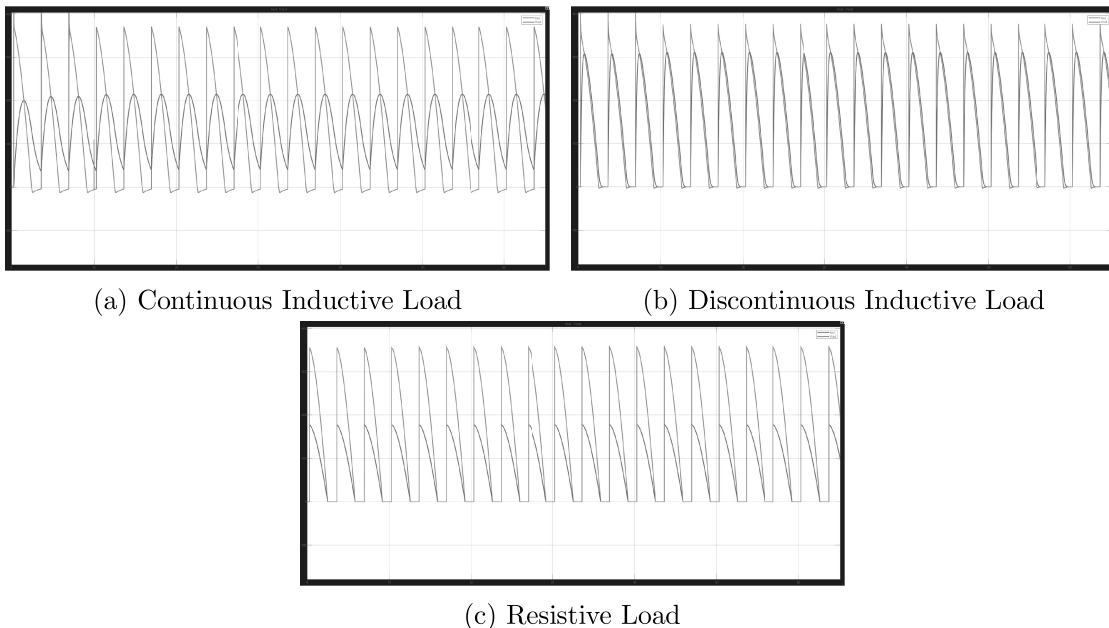


Figure 6: Output Voltage and Current for different loads at $\alpha = 100^\circ$

4 Conclusion

The Vienna Rectifier model achieves reliable and stable AC–DC conversion by utilizing its diode-based rectification stage, midpoint clamping arrangement, and regulated switching operation. The system is supplied by a balanced three-phase source, where each phase reaches a peak value of 311.127 V. These inputs first pass through the 1 mH series inductors, which shape the line currents before they enter the rectifier structure. The resulting DC output increases smoothly from startup and settles within the 402–452.6 V range, exhibiting a switching-related ripple of roughly 50 V. Likewise, the output current ramps upward in a controlled manner, reaching approximately 429–430.5 A under the applied load of $R = 1\ \Omega$ and $L = 8\ \text{H}$, confirming effective DC-link stabilization. Analysis of the diode conduction behavior shows correct rectification, with each diode path conducting in alternating half-cycles and functioning in coordination with the midpoint and switching stage. These observations verify that the Vienna Rectifier delivers a high-quality, low-ripple DC output, making it a strong candidate for high-power fast-charging and other advanced power-conversion applications.

References

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